

KEITHLEY 2000: replacing the VFD with a TFT Color Display

1. Abstract



This Conversion Guide V 2.0 is an extension of Version 1.0 and applies to the new Hardware V2.0. Please refer to Conversion Guide V1.0 for all other information.

With Hardware V2.0, it is now possible to choose between two options: either populate only the STM32 section and utilize the EastRising LT7680 Graphics Acceleration Control Board (ER-PCBA5981), or fully populate the board not needing to use the ER-PCBA5981 module any more. The latter approach makes sense only if the LT7680A-R component can be sourced at a reasonable price - which is currently not the case in Europe.

2. Resources

See Conversion Guide V1.0.

3. Mechanical

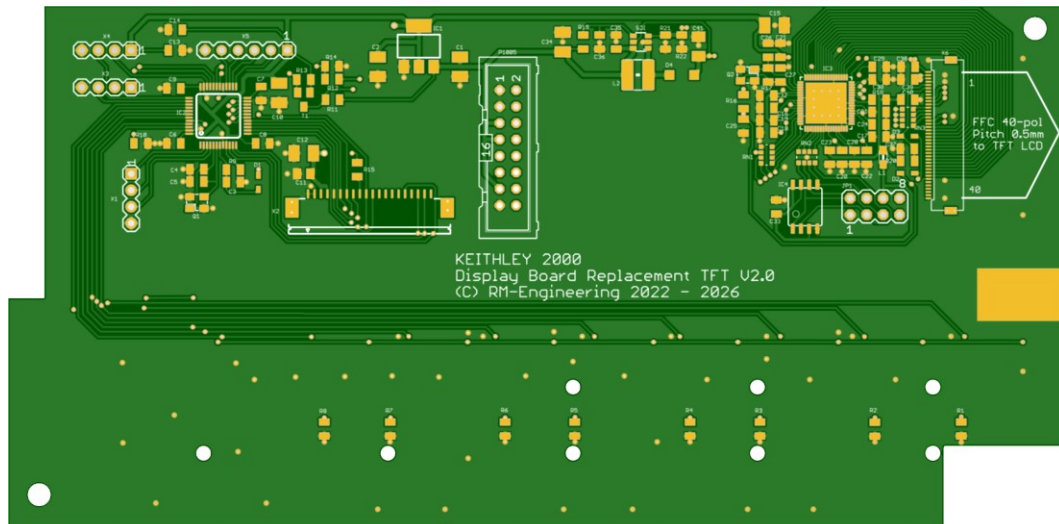
See Conversion Guide V1.0.

KEITHLEY 2000: replacing the VFD with a TFT Color Display

4. Electronic Hardware

The expanded schematic of Hardware V2.0 is based on the reverse engineering of the EastRising board, incorporating several optimizations.

The PCB “KEITHLEY 2000 Display Board Replacement TFT V2.0” has not been manufactured until today because I personally have no need to do this. The PCB design is thoroughly tested and should be error free. Here only an unpopulated picture generated by a Gerber Viewer can be shown.



You can either completely assemble the PCB V2.0 in accordance with the associated Component Placement and the Bill of Material (see Appendix) or just populate the left side as shown in the Conversion Guide V1.0 and place the EastRising Graphics Control Board on the right side using nonconductive double sided mounting tape.

5. Front Assembly

See Conversion Guide V1.0.

6. Programming the Flash Memory and the STM32 Firmware

See Conversion Guide V1.0, note the different pin arrangement of the Flash Programming Connector.

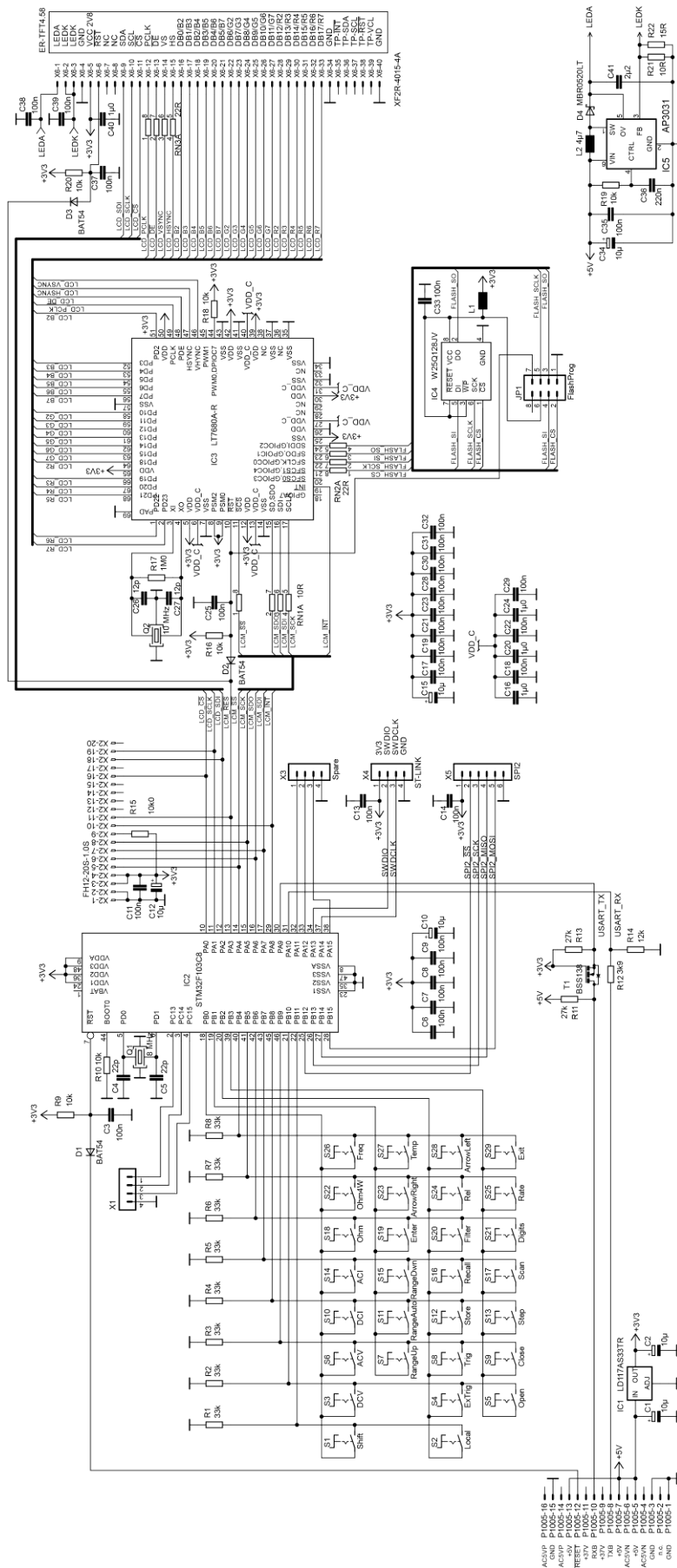
7. Author's Comment

If anyone in Europe happens to have the new PCB V2.0 manufactured, I would be grateful to receive a sample. In that case, I would appreciate it if you could get in touch.

The complete EastRising LT7680 Graphics Acceleration Control Board (ER-PCBA5981) with all its components costs approximately US\$10. Currently, I am unable to source an LT7680A-R chip on its own for less than US\$20. If anyone in Europe has found a better source for this IC priced under US\$10, please let me know.

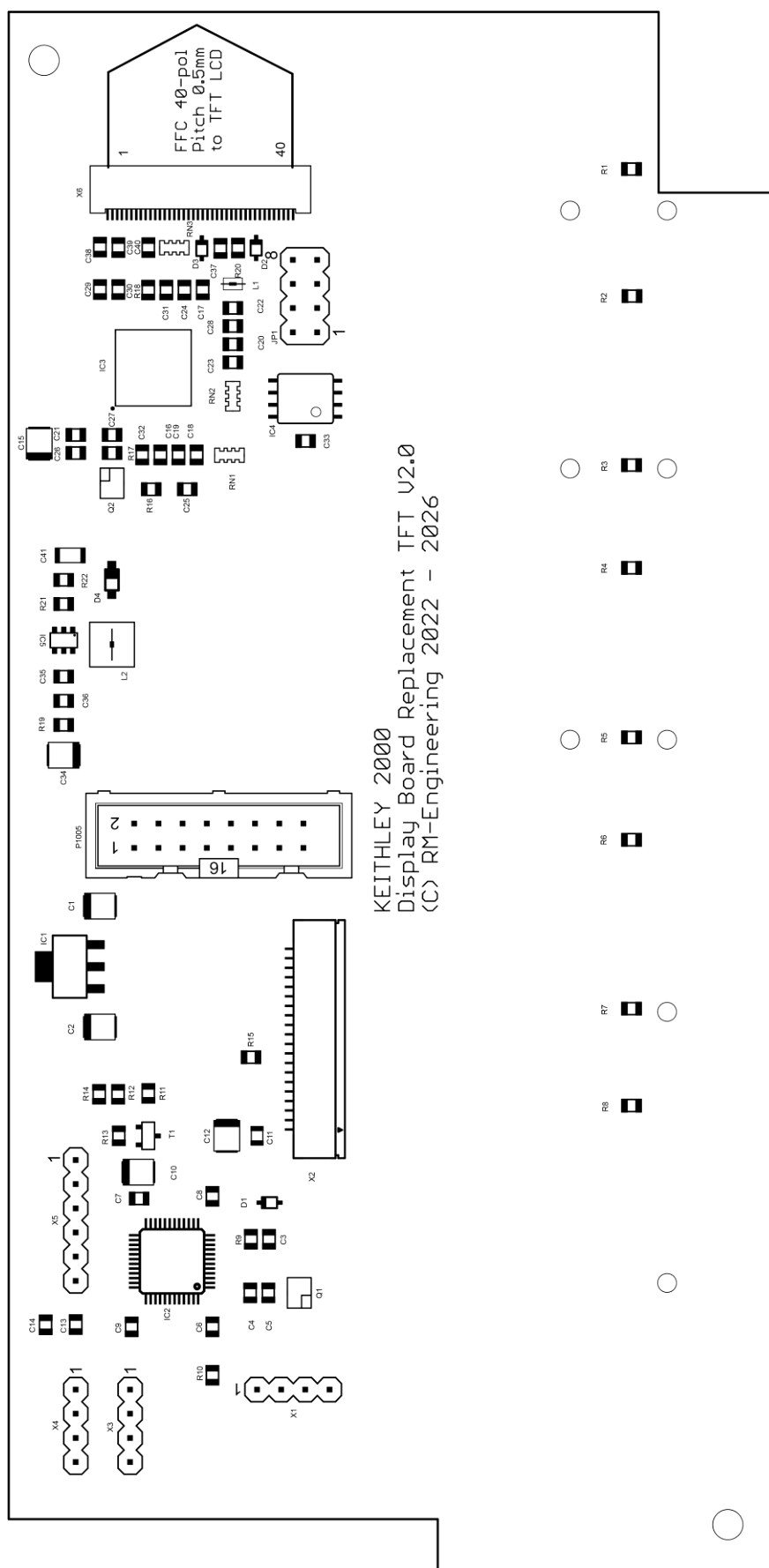
KEITHLEY 2000: replacing the VFD with a TFT Color Display

Appendix 1: Schematic



KEITHLEY 2000: replacing the VFD with a TFT Color Display

Appendix 2: Component Placement



KEITHLEY 2000: replacing the VFD with a TFT Color Display

Appendix 3: Bill of Material

Part	Value	Package	Description	Part	Value	Package	Description
R1	33k	805	Thick Film SMD Resistor	D1	BAT54J	SOD323	Schottky Diode
R2	33k	805	Thick Film SMD Resistor	D2	BAT54J	SOD323	Schottky Diode
R3	33k	805	Thick Film SMD Resistor	D3	BAT54J	SOD323	Schottky Diode
R4	33k	805	Thick Film SMD Resistor	D4	MBS130LT3	SMB/403A	Schottky Power Rectifier
R5	33k	805	Thick Film SMD Resistor	T1	BSS138	SOT23	N-CHANNEL MOS FET
R6	33k	805	Thick Film SMD Resistor	IC1	LD1117AS33TR	SOT223	Low drop fixed positive voltage regulator 1 A
R7	33k	805	Thick Film SMD Resistor	IC2	STM32F103C8	LQFP48	Medium-density performance line ARM®-based 32-bit MCU
R8	33k	805	Thick Film SMD Resistor	IC3	LT7680A-R	QFN-68	High Performance TFT-LCD Graphics Contoller, Levvetop
R9	10k	805	Thick Film SMD Resistor	IC4	W25Q128JV	SOIC-8 208-mil	3V 128M-Bit Serial Flash Memory, Winbond
R10	10k	805	Thick Film SMD Resistor	IC5	AP3031	SOT23-6	White LED Step-Up Converter, Diodes Incorporated
R11	27k	805	Thick Film SMD Resistor	Q1	8 MHz	3225 SMT	Quarz Xtal 3.2mm x2.5mm
R12	3k9	805	Thick Film SMD Resistor	Q2	10 MHz	3225 SMT	Quarz Xtal 3.2mm x2.5mm
R13	27k	805	Thick Film SMD Resistor	L1	Ferrit	805	SMD EMI Suppression Ferrite Bead
R14	12k	805	Thick Film SMD Resistor	L2	4µ7	1616 FPS	SMD-Power Inductor
R15	10k	805	Thick Film SMD Resistor	P1005	Front Panel Connector	16 Pin Male Connector	IDC male 2.54mm 2 Row 16 Contacts upright
R16	10k	805	Thick Film SMD Resistor	JP1	MA04-2	8x2.54	Flash Programm Connector
R17	1M0	805	Thick Film SMD Resistor	X1	MA04-1	4x2.54	do not populate
R18	10k	805	Thick Film SMD Resistor	X2	FPC connector	FPC 20 Pin 1.0mm Pitch,Bottom Contact	do not populate
R19	10k	805	Thick Film SMD Resistor	X3	MA04-1	4x2.54	do not populate
R20	10k	805	Thick Film SMD Resistor	X4	ST-LINK	4x2,54	ST-Link Connector, 4 Pin Header
R21	10R	805	Thick Film SMD Resistor	X5	MA06-1	6x2.54	do not populate
R22	15R	805	Thick Film SMD Resistor	X6	FPC Connector, XF2R-4015-4A	FPC 40 Pin 0.5mm Pitch	FPC connector to LCD Display, Bottom Contact
RN1	10R	1206	Array SMD Chip Resistor	Display	320x960 IPS TFT LCD	ER-TFT4.58-1	EastRising Bar Type 4.58 inch IPS TFT Display with <u>NO Touch Panel</u>
RN2	22R	1206	Array SMD Chip Resistor				
RN3	22R	1206	Array SMD Chip Resistor				
C1	10µ/20V	SMB	Tantal SMD Capacitor				
C2	10µ/10V	SMB	Tantal SMD Capacitor				
C3	100n	805	Ceramic SMD Capacitor				
C4	22p	805	Ceramic SMD Capacitor				
C5	22p	805	Ceramic SMD Capacitor				
C6	100n	805	Ceramic SMD Capacitor				
C7	100n	805	Ceramic SMD Capacitor				
C8	100n	805	Ceramic SMD Capacitor				
C9	100n	805	Ceramic SMD Capacitor				
C10	10µ/10V	SMB	Tantal SMD Capacitor				
C11	100n	805	Ceramic SMD Capacitor				
C12	10µ/10V	SMB	Tantal SMD Capacitor				
C13	100n	805	Ceramic SMD Capacitor				
C14	100n	805	Ceramic SMD Capacitor				
C15	10µ/10V	SMB	Tantal SMD Capacitor				
C16	1µ0	805	Ceramic SMD Capacitor				
C17	100n	805	Ceramic SMD Capacitor				
C18	100n	805	Ceramic SMD Capacitor				
C19	100n	805	Ceramic SMD Capacitor				
C20	1µ0	805	Ceramic SMD Capacitor				
C21	100n	805	Ceramic SMD Capacitor				
C22	100n	805	Ceramic SMD Capacitor				
C23	100n	805	Ceramic SMD Capacitor				
C24	1µ0	805	Ceramic SMD Capacitor				
C25	100n	805	Ceramic SMD Capacitor				
C26	12p	805	Ceramic SMD Capacitor				
C27	12p	805	Ceramic SMD Capacitor				
C28	100n	805	Ceramic SMD Capacitor				
C29	100n	805	Ceramic SMD Capacitor				
C30	100n	805	Ceramic SMD Capacitor				
C31	100n	805	Ceramic SMD Capacitor				
C32	100n	805	Ceramic SMD Capacitor				
C33	100n	805	Ceramic SMD Capacitor				
C34	10µ/20V	SMB	Tantal SMD Capacitor				
C35	100n	805	Ceramic SMD Capacitor				
C36	220n	805	Ceramic SMD Capacitor				
C37	100n	805	Ceramic SMD Capacitor				
C38	100n	805	Ceramic SMD Capacitor				
C39	100n	805	Ceramic SMD Capacitor				
C40	1µ0	0805	Ceramic SMD Capacitor				
C41	2µ2/20V	1206	Ceramic SMD Capacitor				